Rule-Constrained Fine-Tuning and Task Transfer in Large Language Models: Behavioral Evaluation and Causal Analysis of Attention Heads

Abstract

Fine-tuning can produce behavior beyond the source task, but whether it transfers reusable decision structures or yields reliable rule following remains unclear. We examine this question using Dou Dizhu, a rule-rich Chinese climbing card game, as a structured source domain. We compare each Base checkpoint with its Low-Rank Adaptation (LoRA) counterpart for three open-weight models (Qwen2.5-7B, Llama3.1-8B, and DeepSeek-R1-Distill-Llama-8B). On a self-constructed 300-item benchmark spanning Hierarchy Logic, Pattern Recognition, and Counter Logic, overall accuracy increases for all three models, but category-level effects are selective and model-dependent: Qwen2.5-7B improves substantially on Hierarchy Logic and Pattern Recognition while declining on Counter Logic. For Qwen2.5-7B, we introduce the best constraint delta margin (BCDM), which measures the change in relative log-probability between a prohibited action and the best-scoring legal alternative. Mean BCDM increases substantially after adaptation, yet greedy decoding selects the prohibited action in every tested case, revealing a clear gap between probabilistic preference and action-level compliance. Complete attention-head ablation shows heterogeneous causal changes concentrated in a small high-effect subset. Joint ablation of the highest-ranked heads causes greater BCDM degradation than random controls, with non-monotonic effects across intervention sizes. These interventions identify causal support for constraint-sensitive preference shifts, not the mechanism of benchmark gains. Taken together, the results establish fine-tuning as a selective be-

havioral and mechanistic reorganization rather than a uniform capability gain: it improves some structurally related reasoning, concentrates causal dependence for preference adaptation, and can simultaneously worsen rule-compliant generation. Preference adaptation and executable compliance are therefore distinct outcomes, and success on one cannot serve as evidence for the other.
Keywords: Rule-constrained fine-tuning, Task transfer, Constraint-sensitive decision making, Mechanistic interpretability, Attention-head ablation

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1. Introduction

Large language models (LLMs) can acquire new task behaviors through fine-tuning, allowing pretrained models to adapt to specific domains, instructions, and decision requirements [1, 2]. Instruction tuning has further demonstrated that supervision across collections of tasks can improve performance beyond the exact training examples observed during optimization [3, 4]. Parameter-efficient approaches such as Low-Rank Adaptation (LoRA) have made this type of adaptation increasingly practical by enabling substantial behavioral changes without updating all model parameters [5]. However, an important question remains unclear: when a model learns a structured decision policy through fine-tuning, do the resulting changes represent transferable decision structures, constraint-sensitive behavioral adaptation, or merely surface-level improvements specific to the training domain?

This question is particularly important for decision tasks in which the preferred action depends on changing conditions. In realistic scenarios, effective decision making requires more than mapping an input state to a fixed output. A previously preferred action may become invalid after an additional rule, prohibition, or contextual constraint is introduced, requiring the model to suppress that action and select an alternative. We refer to this setting as *rule-constrained fine-tuning*. Compared with ordinary task fine-tuning, rule-constrained examples explicitly supervise how decisions should change when the admissible action space is modified, providing a controlled environment for studying conditional decision behavior.

A key challenge in studying such transfer is separating task-specific surface forms from underlying decision operations. Improvements on conventional benchmarks may reflect better instruction following, memorized task knowledge, or shared linguistic patterns, rather than transfer of reusable reasoning structures. Therefore, a stricter evaluation should examine whether behaviors learned in one structured domain remain effective when the original terminology and context are replaced by different but structurally related tasks.

To investigate this problem, we use Dou Dizhu as a structured source domain. The choice of Dou Dizhu is not intended to study game playing itself, but to provide a compact environment that combines several properties relevant to controlled decision analysis. Unlike many classical rule-based environments that focus primarily on fixed planning or adversarial search, Dou Dizhu contains hierarchical relations, combinatorial patterns, legal action constraints, and conditional action selection within a unified decision process. These characteristics make it suitable for studying whether fine-tuning can transfer abstract decision operations beyond source-domain surface forms.

We construct a mixed fine-tuning corpus combining general instruction data with Dou Dizhu decision examples. In a subset of decision examples, the action that would otherwise be preferred is explicitly prohibited, requiring the model to select a legal alternative. This training setting allows us to investigate not only whether models improve on related reasoning tasks, but also whether fine-tuning changes how models redistribute preferences when previously favored actions become unavailable.

We study this problem from three complementary perspectives. First, we evaluate whether decision structures learned from the source domain transfer to tasks with different surface forms. To this end, we introduce ACL

Bench (Abstracted Cognitive Logic Benchmark)[1], a 300-item benchmark that abstracts source-domain decision operations into three reasoning categories: Hierarchy Logic, Pattern Recognition, and Counter Logic. ACL Bench removes domain-specific terminology while preserving underlying structural operations, enabling evaluation of whether fine-tuning is associated with transfer at the level of reasoning structure.

Second, we examine how fine-tuning changes model behavior when explicit constraints invalidate previously preferred actions. Rather than measuring only whether a prohibited action becomes less likely, we focus on whether the model reallocates preference toward an appropriate legal alternative. We therefore construct controlled constraint-sensitive decision probes and introduce the best constraint delta margin, a log-probability-based metric for measuring preference shifts under explicit constraints.

Third, behavioral evaluation cannot reveal how fine-tuning changes the internal components supporting the observed behavior. We therefore perform causal intervention analysis to examine how functional dependence across attention heads changes after adaptation. This analysis is intended to characterize the internal changes associated with constraint-sensitive behavior rather than directly explain the source of cross-task transfer.

Our experiments reveal three main findings. First, rule-constrained fine-tuning produces cross-task transfer on ACL Bench across multiple open-weight models, but the effects vary substantially across reasoning categories and model families, indicating selective rather than uniform capability improvement. Second, fine-tuning substantially strengthens constraint-sensitive preference reallocation, while the resulting preference changes do not necessarily translate into improved rule-compliant generation. Third, causal intervention analysis reveals sparse and heterogeneous changes in attention-head contributions after fine-tuning, suggesting that behavioral adaptation is asso-

[1]The benchmark implementation and interactive visualization are available at https://github.com/farshelina/rule-constrained-task-transfer.

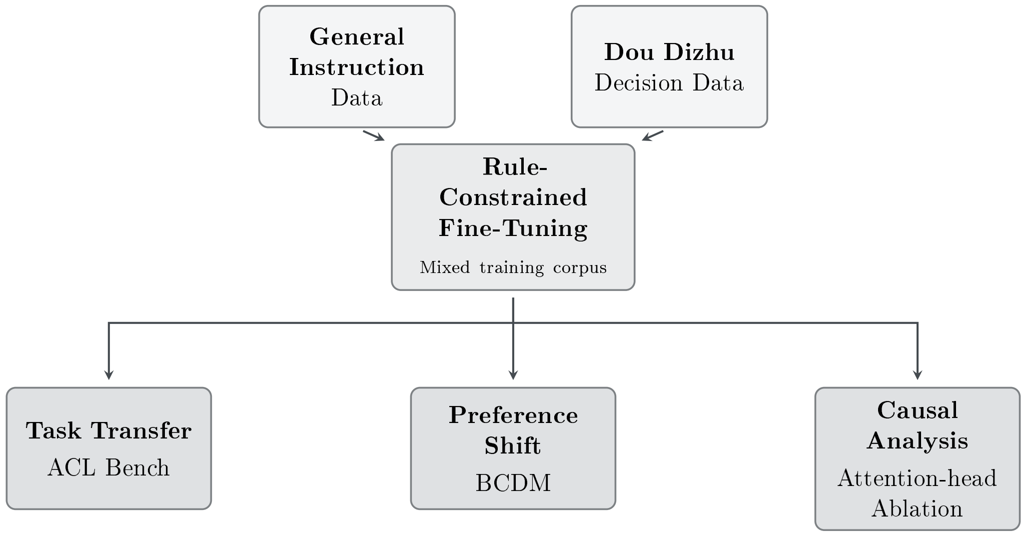


Figure 1: Overview of the proposed rule-constrained fine-tuning framework. The framework integrates structured domain adaptation, cross-task transfer evaluation, constraint-sensitive behavioral analysis, and causal investigation of internal model changes.

ciated with selective changes in functional dependence rather than uniform modification across the attention system.

The main contributions of this work are summarized as follows:

- We establish a controlled framework for studying whether decision behaviors learned through rule-constrained fine-tuning transfer beyond a structured source domain.
- We introduce ACL Bench and the best constraint delta margin to separately evaluate structural task transfer and constraint-sensitive preference reallocation.
- We use causal interventions on attention heads to characterize changes in functional dependence associated with constraint-sensitive decision behavior after fine-tuning.

The remainder of this paper is organized as follows. Section 2 reviews related work on fine-tuning transfer, constraint following, and mechanistic interpretability. Section 3 introduces the proposed framework, benchmark construction, and evaluation metrics. Section 4 describes the experimental settings. Section 5 presents the behavioral and causal analysis results. Section 6 discusses limitations and implications. Finally, Section 7 concludes the study.

2. Related Work

2.1. Fine-Tuning and Cross-Task Generalization

Instruction tuning has shown that supervision on one collection of tasks can improve performance beyond the exact training instances. FLAN demonstrated substantial zero-shot improvements on previously unseen tasks after instruction tuning across heterogeneous datasets [3], while T0 showed that multitask prompted training can support generalization to held-out task formulations [6]. Subsequent work further demonstrated that increasing task diversity, model scale, and reasoning-oriented supervision can improve generalization across broad benchmark collections [4].

These studies establish that fine-tuning can produce behavioral effects beyond the immediate training data. However, generalization across heterogeneous benchmark collections does not necessarily reveal what has transferred. Improved performance may result from better instruction interpretation, broader task exposure, shared linguistic patterns, or reusable reasoning operations.

Broad benchmarks such as MMLU and BIG-Bench evaluate language models across diverse reasoning and knowledge domains [7, 8]. Our setting addresses a narrower question. Rather than testing general competence across unrelated tasks, we study whether recurring decision operations learned from a single structured source domain transfer when the original terminology and context are removed. ACL Bench therefore focuses on structurally related transfer through Hierarchy Logic, Pattern Recognition, and Counter Logic.

This distinction also motivates category-level evaluation. Uniform improvement across reasoning categories would be consistent with a broad capability gain, whereas selective changes across structurally different categories provide a more fine-grained view of what fine-tuning transfers. Low-Rank Adaptation (LoRA) makes this question particularly relevant because substantial behavioral adaptation can be achieved without full-model retraining [5].

2.2. Constraint Following and Rule-Aware Decision Making

Recent work on instruction following has increasingly emphasized explicit constraint satisfaction rather than general response quality. IFEval introduced a reproducible benchmark based on objectively verifiable instructions, including requirements on response length, keywords, and formatting [9]. FollowBench extended this direction to multiple categories of fine-grained constraints and progressively more complex combinations of requirements [10]. These studies show that satisfying explicit constraints constitutes a distinct challenge even for strong language models.

The rule-constrained setting considered in this work differs from conventional response-level constraint evaluation. Rather than specifying properties that the final response must satisfy, our constraints modify the admissible decision space. A previously preferred action can become explicitly prohibited, requiring the model to revise its preference and select a legal alternative.

This distinction motivates our emphasis on preference reallocation. Suppressing the probability of a prohibited action does not necessarily imply that probability has been redirected toward an appropriate alternative. The best constraint delta margin is therefore designed to characterize the relative shift between a forbidden action and the strongest legal alternative. In this respect, our work complements existing constraint-following benchmarks by examining how fine-tuning changes decision preference when the set of admissible actions changes.

2.3. Mechanistic Interpretability and Fine-Tuning

Behavioral evaluation alone cannot determine which internal model components become more or less important after adaptation. Mechanistic interpretability addresses this limitation through causal interventions and circuit-level analyses that identify components functionally involved in specific behaviors [11–13].

Prior work has shown that individual attention heads can implement behaviorally meaningful computations. Olsson et al. identified induction heads associated with sequence-completion behavior and provided causal evidence for their functional role [14]. Intervention-based approaches have also demonstrated that causal perturbations can reveal internal components that are important for particular model outputs or computations [12, 13]. These findings motivate the use of causal intervention rather than attention magnitude or correlation alone when evaluating functional importance.

More recent studies have examined how fine-tuning changes internal mechanisms. Prakash et al. found that fine-tuned models performing entity tracking largely reused mechanisms already present in the base model, with improved performance associated with strengthening of pre-existing circuitry [15]. Chhabra et al. further reported localized amplification, disruption, and recovery of task-relevant mechanisms under different fine-tuning conditions [16]. Together, these results suggest that fine-tuning can selectively modify the functional importance of existing model components rather than uniformly altering the network.

Our mechanistic analysis follows this intervention-based perspective but focuses on constraint-sensitive decision behavior. We compare attention-head causal contributions before and after fine-tuning and subsequently test whether the heads showing the largest changes form a functionally distinctive subset. Importantly, this analysis concerns the internal support for constraint-sensitive preference adaptation; it does not by itself establish the causal mechanism underlying ACL Bench transfer.

Table 1: Comparison between representative related research directions and the present study.

Research direction	**Representative work**	**Relation to the present study**
Cross-task generalization	FLAN [3], T0 [6], FLAN scaling [4]	Studies generalization across heterogeneous tasks; our work isolates transfer of structurally related decision operations from a single structured source domain.
Constraint following	IFEval [9], FollowBench [10]	Evaluates compliance with directly verifiable response requirements; our setting changes the admissible action space and examines preference reallocation after a preferred action is prohibited.
Mechanistic interpretability	Olsson et al. [14], Meng et al. [12], Conmy et al. [13]	Uses causal intervention to identify functionally important components; our work compares attention-head causal contributions before and after rule-constrained fine-tuning.
Mechanisms after fine-tuning	Prakash et al. [15], Chhabra et al. [16]	Studies reuse and modification of internal mechanisms after adaptation; our work focuses specifically on causal dependence associated with constraint-sensitive preference changes.

Overall, the present study lies at the intersection of three research directions: cross-task transfer after fine-tuning, explicit constraint-sensitive decision making, and causal analysis of fine-tuned models. The central distinction is that these three levels are evaluated within the same controlled source-domain setting while remaining analytically separate: ACL Bench measures structural task transfer, constraint-sensitive probes measure preference reallocation, and causal intervention examines the internal components supporting the latter behavior.

3. Methodology

Our study investigates rule-constrained fine-tuning at three complementary levels: source-domain training, cross-task behavioral transfer, and internal causal dependence. We first construct a mixed fine-tuning corpus in which a subset of structured decision examples contains explicit prohibitive constraints. We then evaluate cross-task transfer using ACL Bench, which preserves selected structural operations from the source domain while changing their surface form. To characterize constraint-sensitive preference reallocation directly, we construct controlled decision probes and introduce the Best Constraint Delta Margin (BCDM). Finally, we perform complete single-head causal ablation and joint Top-k interventions to examine how fine-tuning changes the functional importance of individual attention heads.

3.1. Dataset Construction

3.1.1. Mixed Fine-Tuning Corpus

We use Dou Dizhu as a structured rule-based decision environment. The purpose of the source domain is not to optimize game-playing performance itself, but to provide repeated examples of hierarchical comparison, structured pattern recognition, legal-action selection, and conditional decision adjustment.

The final fine-tuning corpus contains 55,900 instruction–response examples, consisting of 50,400 general instruction examples and 5,500 Dou Dizhu decision examples. The general instruction data are included to preserve broad instruction-following behavior, while the domain-specific examples expose the model to the structured decision operations of interest.

All fine-tuned models are adapted using Low-Rank Adaptation (LoRA) [5], allowing task-specific behavior to be learned while keeping the pretrained model parameters frozen.

3.1.2. Ordinary and Rule-Constrained Decision Examples

We distinguish between ordinary decision examples and rule-constrained decision examples. An ordinary decision example can be represented as $x \to a^*$, where x denotes the current decision state and a^* is the preferred legal action under the original rules. For a rule-constrained example, an additional condition c explicitly prohibits an action that would otherwise be preferred, yielding $(x, c) \to a^c$, where a^c is a legal alternative under the modified action space.

The final rule-constrained corpus contains 851 constraint-injected instructions. Among these examples, 646 require a change in the target output after the additional constraint is introduced. This construction exposes the model to cases in which a previously preferred action becomes unavailable and the decision policy must therefore be revised rather than simply reproduced.

Conceptually, the supervision changes from selecting the preferred action under the original candidate set to selecting the preferred legal action after prohibited candidates are removed:

$$\begin{aligned} a^* &= \arg\max_{a \in \mathcal{A}} P(a \mid x), \\ a^c &= \arg\max_{a \in \mathcal{A} \setminus \mathcal{F}_c} P(a \mid x, c), \end{aligned} \tag{1}$$

where $\mathcal{A}$ denotes the original candidate-action set and $\mathcal{F}_c$ denotes the set of actions prohibited by constraint c.

The model trained on the rule-constrained corpus is referred to as **LoRA-v2** throughout the paper. An ordinary version of the mixed fine-tuning corpus without the special constraint intervention is retained as LoRA-v1. However, the primary results in the present study focus on the Base and LoRA-v2 comparison. A strictly matched LoRA-v1 versus LoRA-v2 experiment is reserved for isolating the incremental contribution of the constraint-injected examples and is not included in the current main results.

3.2. ACL Bench Construction

To evaluate whether behavioral patterns acquired in the source domain transfer to tasks with different surface forms, we construct the Abstracted Cognitive Logic Benchmark (ACL Bench).

ACL Bench contains 300 four-choice multiple-choice questions divided equally into three reasoning categories:

- **Hierarchy Logic**: reasoning over explicitly ordered hierarchical relations, including comparison and rule-dependent access or override relations;
- **Pattern Recognition**: identifying, extending, or distinguishing structured patterns, including sequence completion and outlier recognition;
- **Counter Logic**: selecting the minimum candidate that strictly exceeds a specified threshold, including cases in which no candidate satisfies the condition.

Each category contains 100 questions. The benchmark is designed to preserve abstract operations that recur in the source decision environment while removing direct dependence on Dou Dizhu-specific terminology. Consequently, successful performance does not require knowledge of card-game rules, card names, or source-domain output templates.

The three categories capture different structural relationships between the source and target tasks. Hierarchy Logic abstracts ordered rank relations, Pattern Recognition captures recurring structural regularities, and Counter Logic abstracts conditional selection relative to a comparison threshold.

The benchmark generation procedure uses a fixed random seed of 42. Answer options are shuffled during item construction, and the complete benchmark is additionally shuffled before export. Several quality-control procedures are applied during benchmark construction. Hierarchy Logic items are checked for subject–relation alignment, Pattern Recognition items are checked to prevent duplicated answer options, and Counter Logic prompts are revised to ensure that the comparison condition is expressed consistently. Additional manual and programmatic checks are used to remove typographical errors, rare or unintended lexical artifacts, and context–answer mismatches.

For a model M, category-level accuracy is defined as

$$\mathrm{Acc}_c(M) = \frac{1}{N_c} \sum_{i=1}^{N_c} \mathbb{I}\left[\hat{y}_i^M = y_i\right], \tag{2}$$

where $N_c = 100$ for each category, $\hat{y}_i^M$ is the answer predicted by model M, and y_i is the correct answer. Overall ACL Bench accuracy is computed over all 300 questions:

$$\mathrm{Acc}_{\mathrm{ACL}}(M) = \frac{1}{300} \sum_{i=1}^{300} \mathbb{I}\left[\hat{y}_i^M = y_i\right]. \tag{3}$$

The purpose of ACL Bench is therefore not to measure unrestricted general reasoning ability, but to test whether fine-tuning is associated with transfer across tasks that share selected structural operations with the source domain.

3.3. Constraint-Sensitive Preference-Shift Metric

ACL Bench evaluates cross-task transfer, but does not directly measure whether the model has learned to revise its decision preference when an explicit prohibition is introduced. To examine this behavior, we construct 60 controlled constraint-sensitive decision probes covering six decision scenarios. The paired design follows the broader behavioral-testing and contrast-set paradigm, in which targeted test cases or small but meaningful input changes are used to probe whether model decisions respond to the intended condition [17, 18].

Each probe contains two closely matched conditions:

1. an unconstrained condition containing the original decision state;

2. a constrained condition in which an additional rule explicitly prohibits an otherwise preferred action.

For probe i, let f_i denote the forbidden action and let $\mathcal{L}_i$ denote the set of legal alternatives. For a model M, the candidate-level score of action a under context x is defined as $s_M(a \mid x) = \log P_M(a \mid x)$.

We first compute the relative margin between the strongest legal alternative and the forbidden action in the unconstrained condition, and then recompute the same margin after constraint c_i is introduced:

$$m_i^{\text{clean}} = \max_{a \in \mathcal{L}_i} s_M(a \mid x_i) - s_M(f_i \mid x_i), \tag{4}$$

$$m_i^{\text{constraint}} = \max_{a \in \mathcal{L}_i} s_M(a \mid x_i, c_i) - s_M(f_i \mid x_i, c_i). \tag{5}$$

We define the Best Constraint Delta Margin (BCDM) for probe i as

$$\text{BCDM}_i = m_i^{\text{constraint}} - m_i^{\text{clean}}. \tag{6}$$

The model-level preference-shift score is the mean across all $N = 60$ probes:

$$S(M) = \frac{1}{N} \sum_{i=1}^{N} \text{BCDM}_i. \tag{7}$$

A positive BCDM indicates that introducing the constraint shifts relative model preference away from the forbidden action and toward the strongest legal alternative. This metric differs from simple forbidden-action suppression: a model may reduce the probability of a prohibited action without increasing the probability of an appropriate legal response. By explicitly comparing the forbidden action with the strongest legal alternative, BCDM measures *preference reallocation* rather than suppression alone.

3.4. Attention-Head Causal Ablation

Behavioral differences between the Base and fine-tuned models do not reveal which internal components become more or less important after adaptation. Prior work has used interventions on neurons and attention heads to identify components that are causally implicated in model behavior [13, 19, 20]. We therefore perform causal intervention at the level of individual attention heads. Head-removal studies further show that functional contribution can be highly uneven across heads [21], motivating a complete scan rather than an assumption of uniform head importance.

The primary mechanistic analysis is conducted on Qwen2.5-7B, which contains 28 Transformer layers with 28 attention heads per layer, for a total of $28 \times 28 = 784$ attention heads. For each head (l, h), we perform single-head zero ablation during the forward pass while leaving the remaining attention heads and model components unchanged. The resulting model is reevaluated on the same 60 constraint-sensitive probes using BCDM.

For a model M, let $S_{\text{original}}(M)$ denote its preference-shift score without intervention and $S_{\text{ablated}}(M, l, h)$ the score after ablating attention head (l, h). We define the causal effect of head (l, h) as

$$CE_M(l, h) = S_{\text{original}}(M) - S_{\text{ablated}}(M, l, h). \tag{8}$$

A positive value indicates that removing the head decreases constraint-sensitive preference reallocation, implying that the head makes a positive functional contribution to the measured behavior. A negative value indicates that removing the head increases the score, suggesting that the head does not positively support the measured behavior under the current metric.

To quantify how fine-tuning changes the causal importance of each head, we define the Base-to-LoRA causal-change measure ΔCE as

$$\Delta CE(l, h) = CE_{\text{LoRA-v2}}(l, h) - CE_{\text{Base}}(l, h). \tag{9}$$

A large positive ΔCE indicates that the fine-tuned model becomes more dependent on that attention head for the measured constraint-sensitive preference shift. The complete procedure is applied independently to all 784 attention heads in both the Base and LoRA-v2 models, yielding 784 single-head interventions per model and 1,568 intervention measurements in total. This complete scan allows us to analyze both individual high-impact heads and the global distribution of causal changes across layers and head indices.

3.5. Top-k Causal Validation

Large single-head causal effects do not necessarily imply that the highest-ranked heads form a collectively important functional subset. We therefore perform joint Top-k ablation based on the ΔCE ranking. Let $\mathcal{H}_k$ denote the set of the k heads with the largest positive ΔCE values:

$$\mathcal{H}_k = \mathrm{TopK}_{(l,h)} \left[\Delta CE(l,h)\right]. \tag{10}$$

We evaluate $k \in \{5, 10, 27\}$; the $k = 27$ setting corresponds to the complete set of heads with $\Delta CE > 1$ in the single-head ranking. For each k, all heads in $\mathcal{H}_k$ are ablated simultaneously and the preference-shift score is recomputed. The resulting behavioral degradation is defined as

$$D(\mathcal{H}_k) = S_{\mathrm{original}}(M) - S_{\mathrm{ablated}}(M; \mathcal{H}_k). \tag{11}$$

A larger value indicates greater functional dependence on the removed head set.

The complete causal-intervention procedure, including the two random controls described below, is summarized in Algorithm 1.

To determine whether large Top-k effects are specific to the ΔCE ranking rather than a generic consequence of removing multiple attention heads, we compare the ranked intervention with two controls.

Algorithm 1 Attention-Head Causal Scan and Top-k Validation

Input: Base model M_B, LoRA-v2 model M_L, probe set $\mathcal{P}$, head set $\mathcal{A}$, and $K = \{5, 10, 27\}$

Output: Head-wise ΔCE values and Top-k degradation scores

Compute $S_{\text{original}}(M_B)$ and $S_{\text{original}}(M_L)$ on $\mathcal{P}$
for $M \in \{M_B, M_L\}$ **do**
    for each head $(l, h) \in \mathcal{A}$ **do**
        Zero-ablate head (l, h)
        Evaluate $S_{\text{ablated}}(M, l, h)$ on $\mathcal{P}$
        $CE_M(l, h) \leftarrow S_{\text{original}}(M) - S_{\text{ablated}}(M, l, h)$
    end for
end for
for each head $(l, h) \in \mathcal{A}$ **do**
    $\Delta CE(l, h) \leftarrow CE_{M_L}(l, h) - CE_{M_B}(l, h)$
end for
Rank all heads in descending order of ΔCE
for $k \in K$ **do**
    $\mathcal{H}_k \leftarrow$ the k heads with the largest ΔCE
    Jointly ablate $\mathcal{H}_k$ from M_L
    $D_{\text{Top}}(k) \leftarrow S_{\text{original}}(M_L) - S_{\text{ablated}}(M_L; \mathcal{H}_k)$
    for $r = 0, \ldots, 4$ **do**
        Sample $\mathcal{G}_{k,r}$ uniformly from all 784 heads
        Sample $\mathcal{M}_{k,r}$ with the exact layer counts of $\mathcal{H}_k$
        Compute $D^{(r)}_{\text{Global}}(k)$ and $D^{(r)}_{\text{Matched}}(k)$
    end for
    Average the five Global Random and Matched Random degradation scores
end for

3.5.1. Global Random Control

The **Global Random** control samples k heads without replacement from the complete set of 784 attention heads. This control tests whether removing an arbitrary set of k heads is sufficient to produce degradation comparable to the Top-k intervention.

3.5.2. Layer-Matched Random Control

The **Matched Random** control preserves the exact layer-count distribution of the corresponding Top-k set. For each layer, it randomly samples the same number of heads as appear in the Top-k set for that layer. This controls for the possibility that the observed Top-k effect is explained simply by ablating heads from particularly sensitive layers rather than by the identities of the selected heads themselves.

For both random-control conditions, five independently sampled control sets are generated using random seeds 0, 1, 2, 3, and 4. The reported control effect for each k is the mean behavioral degradation across the five runs. Thus, the joint validation compares the ΔCE-selected Top-k set with both a model-wide random baseline and a layer-matched random baseline.

If the ΔCE ranking identifies a functionally distinctive subset, Top-k ablation should produce greater behavioral degradation than both controls. Importantly, this is a joint causal test rather than an additive combination of single-head effects. In general,

$$CE(h_1, \ldots, h_k) \neq \sum_{i=1}^{k} CE(h_i), \tag{12}$$

because attention heads may interact through redundancy, compensation, or other nonlinear dependencies. The joint-ablation experiment therefore tests the collective functional importance of the selected subset without assuming additive head contributions.

Table 2: Main LoRA fine-tuning configuration.

Hyperparameter	Value
LoRA rank	16
LoRA alpha	32
LoRA dropout	0.05
Learning rate	1×10^{-4}
Training epochs	3
Per-device batch size	4
Maximum sequence length	2048
Learning-rate scheduler	Cosine
Warmup ratio	0.1
Gradient checkpointing	Enabled

4. Experimental Setup

4.1. Base Models

We evaluate three open-weight models: Qwen2.5-7B [22], Llama3.1-8B [23], and DeepSeek-R1-Distill-Llama-8B [24]. For each model, **Base** denotes the checkpoint before task-specific adaptation, and **LoRA-v2** denotes the counterpart fine-tuned on the rule-constrained mixed corpus.

Mechanistic experiments are restricted to Qwen2.5-7B, for which the complete single-head scan and Top-k joint-ablation experiments were performed.

4.2. Fine-Tuning Configuration

All models are adapted using Low-Rank Adaptation (LoRA) [5]. The pretrained parameters remain frozen, while trainable low-rank modules are inserted into the attention projections q_{proj}, k_{proj}, v_{proj}, and o_{proj}, and the feed-forward projections $\mathrm{gate}_{\mathrm{proj}}$, $\mathrm{up}_{\mathrm{proj}}$, and $\mathrm{down}_{\mathrm{proj}}$.

The main configuration is listed in Table 2.

LoRA is applied to both attention and feed-forward projections; the mechanistic analysis focuses on attention heads.

4.3. ACL Bench Evaluation

Cross-task transfer is evaluated on the 300-item ACL Bench introduced in Section 3.2. The benchmark contains 100 four-choice questions in each of three categories: Hierarchy Logic, Pattern Recognition, and Counter Logic.

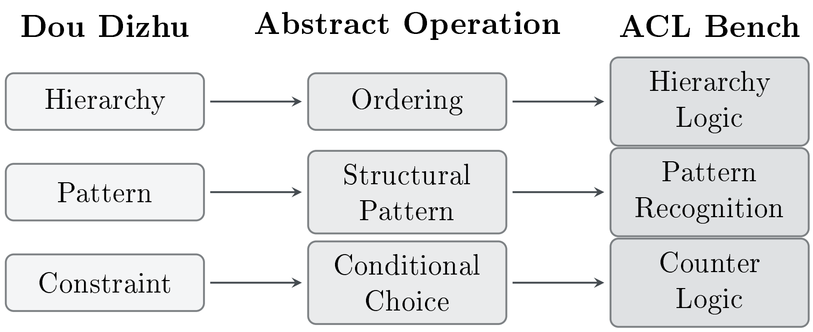


Figure 2: Construction of ACL Bench by abstracting Dou Dizhu decision operations into transferable reasoning categories. The benchmark removes domain-specific surface forms while preserving underlying decision operations.

We report overall and category-level accuracy for each Base/LoRA-v2 pair. Category-level results are retained because fine-tuning effects differ across reasoning types even when overall accuracy improves.

4.4. Constraint-Sensitive Probe Evaluation

Constraint-sensitive behavior is evaluated with 60 controlled probes covering six decision scenarios. Each probe has a clean condition and a constrained condition in which an otherwise preferred action is explicitly prohibited. Evaluation uses seed 42, FP16 inference, SDPA attention, and batch size 1.

The primary comparison is between Qwen2.5-7B Base and Qwen2.5-7B LoRA-v2. We report mean BCDM across all 60 probes and per-scenario scores, using the metric defined in Section 3.3. The six scenarios are reported in Table 5.

These probes are used to measure preference reallocation under explicit prohibition rather than as an independent benchmark.

4.5. Single-Head Causal Analysis

The complete attention-head scan is performed on Qwen2.5-7B, which has 28 Transformer layers and 28 attention heads per layer, for a total of 784 heads. Each head is ablated independently and the model is reevaluated on the same 60 constraint-sensitive probes.

The scan is run for both Base and LoRA-v2, yielding 1,568 single-head intervention evaluations. For each head (l, h), we compute $CE_{\text{Base}}(l, h)$, $CE_{\text{LoRA}}(l, h)$, and $\Delta CE(l, h)$ as defined in Section 3.4. The resulting 28×28 map is used to analyze causal changes and rank heads for joint ablation.

4.6. Top-k Joint-Ablation Evaluation

Joint ablation is evaluated at $k \in \{5, 10, 27\}$. For each k, we compare three conditions: **Top-k**, consisting of the heads with the largest positive ΔCE values; **Global Random**, sampled from all 784 heads; and **Matched Random**, sampled to preserve the exact layer-count distribution of the corresponding Top-k set.

All joint-ablation runs use the same 60 probes, seed 42, FP16 inference, SDPA attention, and batch size 1. For each random-control condition, five independently sampled sets are generated with seeds 0–4, and their degradation scores are averaged.

The two random controls separate generic sensitivity to multi-head removal from effects attributable to layer placement. A larger degradation under Top-k than under both controls indicates that the selected head identities carry additional functional relevance. No additive assumption is made for joint interventions.

4.7. Implementation Environment

Fine-tuning and mechanistic experiments are run under Windows Subsystem for Linux (WSL) on an NVIDIA RTX A4500 GPU with 20 GB of memory. Gradient checkpointing is enabled during LoRA fine-tuning. Mechanistic evaluation uses mixed-precision inference where applicable and batch size 1 to support consistent head-level intervention.

5. Results

This section reports the experimental results at three levels. We first evaluate cross-task transfer on ACL Bench across three model families. We then examine constraint-sensitive preference reallocation using the controlled behavioral probes. Finally, we analyze how fine-tuning changes the causal contribution of individual attention heads and validate the highest-ranked heads through joint Top-k ablation against two random-control conditions.

5.1. Task-Transfer Results

5.1.1. Overall ACL Bench Performance

Table 3 reports accuracy separately for Counter Logic, Hierarchy Logic, and Pattern Recognition.

Table 3: Overall ACL Bench accuracy before and after rule-constrained fine-tuning. Values are percentages; Δ is the Base-to-LoRA-v2 change in percentage points.

Model	Base	LoRA-v2	Δ
DeepSeek-R1-Distill-Llama-8B	28.0	38.0	+10.0
Qwen2.5-7B	35.3	54.0	+18.7
Llama3.1-8B	45.0	58.7	+13.7

All three model families achieve higher overall ACL Bench accuracy after fine-tuning.

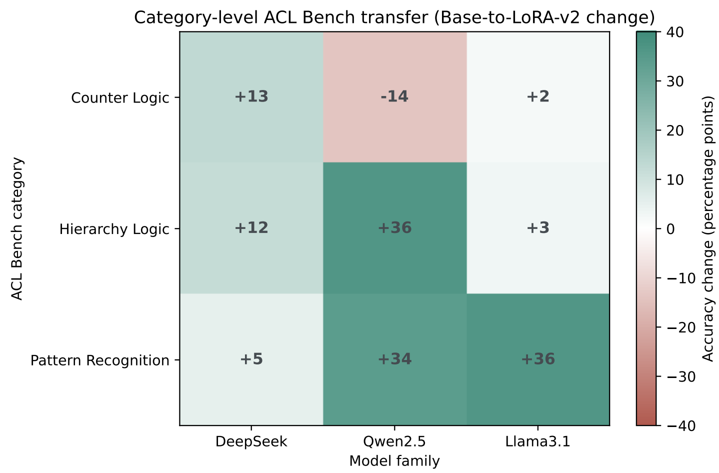
The DeepSeek model improves from 28.0% to 38.0%, a gain of 10.0 percentage points. Qwen2.5-7B shows the largest overall improvement, increasing from 35.3% to 54.0%, while Llama3.1-8B increases from 45.0% to 58.7%.

The consistency of the overall direction across three model families indicates that the behavioral changes induced by the rule-constrained fine-tuning corpus are not limited to a single pretrained model.

However, the category-level results reveal that these gains are highly non-uniform.

5.1.2. Category-Level Transfer

The category-level transfer pattern is summarized in Figure 3.



Model	Setting	Counter	Hierarchy	Pattern
DeepSeek	Base	30	27	27
	LoRA-v2	43	39	32
Qwen2.5	Base	38	34	34
	LoRA-v2	24	70	68
Llama3.1	Base	39	62	34
	LoRA-v2	41	65	70

Figure 3: Category-level ACL Bench transfer after rule-constrained fine-tuning. The heatmap shows Base-to-LoRA-v2 accuracy changes, while the table provides the corresponding category-level scores. The results reveal selective and model-dependent transfer rather than uniform improvement across reasoning categories.

DeepSeek-R1-Distill-Llama-8B shows improvements across all three categories. Counter Logic increases from 30% to 43%, Hierarchy Logic from 27%

to 39%, and Pattern Recognition from 27% to 32%. The improvement is therefore distributed across categories, although its magnitude varies.

Qwen2.5-7B exhibits a substantially different transfer profile. Hierarchy Logic increases from 34% to 70%, and Pattern Recognition increases from 34% to 68%. In contrast, Counter Logic decreases from 38% to 24%.

Llama3.1-8B shows only limited changes in Counter Logic and Hierarchy Logic, increasing from 39% to 41% and from 62% to 65%, respectively. Its largest improvement occurs in Pattern Recognition, which increases from 34% to 70%.

These category-level differences show that the overall gains cannot be interpreted as a uniform increase in general reasoning ability.

Instead, the observed transfer is selective and strongly dependent on both the target reasoning category and the underlying base model.

In particular, Qwen2.5-7B and Llama3.1-8B obtain large improvements in Pattern Recognition, whereas their Counter Logic behavior differs substantially. Qwen2.5-7B improves strongly in Hierarchy Logic but degrades in Counter Logic, while Llama3.1-8B remains comparatively stable in both categories.

The ACL results therefore support a characterization of the fine-tuning effect as structure-dependent and model-dependent cross-task transfer rather than broad, homogeneous reasoning enhancement.

5.2. Constraint-Sensitive Preference-Shift Results

ACL Bench evaluates transfer outside the source domain, but it does not directly measure whether the fine-tuned model has learned to revise its decision preference when an additional rule prohibits an otherwise preferred action.

We therefore evaluate Qwen2.5-7B Base and LoRA-v2 on the 60 controlled constraint-sensitive probes using the best constraint delta margin.

Table 4: Best constraint delta margin on the 60 constraint-sensitive probes. Values are reported as mean ± standard deviation.

Model	BCDM
Qwen2.5-7B Base	1.391 ± 1.379
Qwen2.5-7B LoRA-v2	4.371 ± 0.766

5.2.1. Overall Preference-Reallocation Effect

Table 4 summarizes the overall preference-shift scores.

Mean BCDM increases from 1.391 ± 1.379 for the Base model to 4.371 ± 0.766 after rule-constrained fine-tuning, a gain of approximately 2.981 points. Because BCDM measures the change in relative preference between the forbidden action and the strongest legal alternative, the increase indicates stronger preference reallocation after an explicit prohibition is introduced.

The relatively large standard deviation of the Base model is consistent with the heterogeneous scenario-level behavior reported below: several scenarios produce small mean shifts, whereas `straight_34567` already produces a much larger Base response. The aggregate Base score therefore combines probes with substantially different levels of constraint sensitivity rather than reflecting a uniformly distributed response. LoRA-v2 shows a smaller standard deviation, indicating a more consistent preference shift across the evaluated probes.

Importantly, BCDM should not be interpreted as final-action compliance. It measures how the relative ordering of candidate-action probabilities changes when a prohibition is introduced.

5.2.2. Scenario-Level Results

The 60 probes are divided equally across six controlled scenarios, with 10 probes per scenario. Table 5 reports the corresponding mean scores.

LoRA-v2 achieves a higher constraint-sensitive score in five of the six scenarios.

Table 5: Scenario-level best constraint delta margin for Qwen2.5-7B. Each scenario contains 10 probes.

Scenario	Base	LoRA-v2	Δ
`bomb_9999`	1.898	4.180	+2.282
`pair_88`	0.309	5.284	+4.975
`real_pair_QQ_landlord`	0.155	4.107	+3.952
`rocket_BR`	1.115	5.427	+4.312
`single_A_follow`	0.700	3.252	+2.552
`straight_34567`	4.168	3.977	-0.191

The largest gains are 4.975 for `pair_88`, 4.312 for `rocket_BR`, and 3.952 for `real_pair_QQ_landlord`.

The `straight_34567` scenario differs from the others. The Base model already exhibits a relatively high score of 4.168, while the LoRA-v2 score is slightly lower at 3.977.

The fine-tuning effect is therefore not uniformly positive across every individual scenario.

Nevertheless, the overall mean increases substantially, and five of the six scenario-level comparisons show clear improvement. The results support a cross-scenario enhancement of constraint-sensitive preference reallocation while also indicating that the magnitude of the effect depends on the underlying decision structure.

5.2.3. Preference–Generation Gap

The probability-level improvement does not translate into better greedy-decoding compliance. Under direct generation, the Base model selects the forbidden action in 40 of 60 probes (66.7%) and a legal action in the remaining 20 probes; 10 of these legal outputs match the designated allowed action and 10 are other legal alternatives. LoRA-v2 selects the forbidden action in all 60 probes.

This diagnostic establishes a clear separation between preference adap-

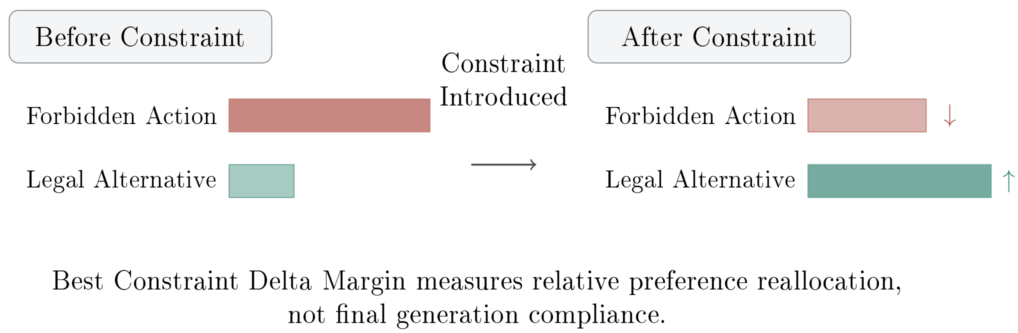


Figure 4: Illustration of constraint-sensitive preference reallocation. The best constraint delta margin measures the relative probability shift from a forbidden action toward the strongest legal alternative after an explicit constraint is introduced.

tation and executable compliance. BCDM improves because the constraint changes the relative margin between forbidden and legal candidates, but the forbidden action can still retain the highest absolute probability and therefore remain the greedy-decoding output. The mechanistic analyses below concern the preference-level effect measured by BCDM, not final-action compliance.

5.3. Attention-Head Causal Results

LoRA-v2 shows substantially stronger constraint-sensitive preference reallocation than the Base model. We next examine whether this difference is accompanied by changes in the functional importance of individual attention heads.

The causal analysis is performed on Qwen2.5-7B.

5.3.1. Exploratory Attention-Based Screening

Before performing the complete intervention scan, we explored whether constraint-related attention heads could be identified through a three-stage attention-based screening procedure consisting of targeting, correlation, and counterfactual filtering.

For the Base model, the screening sequence was 784 $\rightarrow$ 157 $\rightarrow$ 22 $\rightarrow$ 14 candidate heads. The standard LoRA-v2 sequence was 784 $\rightarrow$ 157 $\rightarrow$

$0 \to 0$. Under a relaxed LoRA-v2 setting using 100 samples, the targeting stage retained 235 heads, but none survived the subsequent correlation stage ($784 \to 235 \to 0 \to 0$).

These results indicate that simple attention–behavior correlation does not provide a stable basis for identifying constraint-relevant components after fine-tuning.

Rather than treating the absence of correlated heads as evidence that the model lacks localized functional dependence, we therefore proceed to direct causal intervention.

5.3.2. Complete Single-Head Causal Scan

Qwen2.5-7B contains 28 Transformer layers and 28 attention heads per layer, resulting in 784 attention heads.

Every attention head is independently ablated in both the Base and LoRA-v2 models, yielding 1,568 single-head interventions. For each head, we compute the causal contribution and the Base-to-LoRA change $\Delta CE(l, h)$ defined in Eq. 9.

The full scan reveals a highly heterogeneous distribution.

Most attention heads show relatively small Base-to-LoRA changes, while a much smaller subset exhibits substantially larger positive ΔCE values.

Negative values are also observed, indicating that some heads become less supportive of the measured behavior after fine-tuning or that their removal improves the preference-shift score.

Figure 6 summarizes positive causal contribution across Transformer layers.

The causal changes are therefore not distributed uniformly across the attention system.

Table 6 reports the ten heads with the largest positive ΔCE values.

Several of the strongest causal changes are concentrated in Layer 0. In particular, five of the ten highest-ranked heads are located in the first Transformer layer.

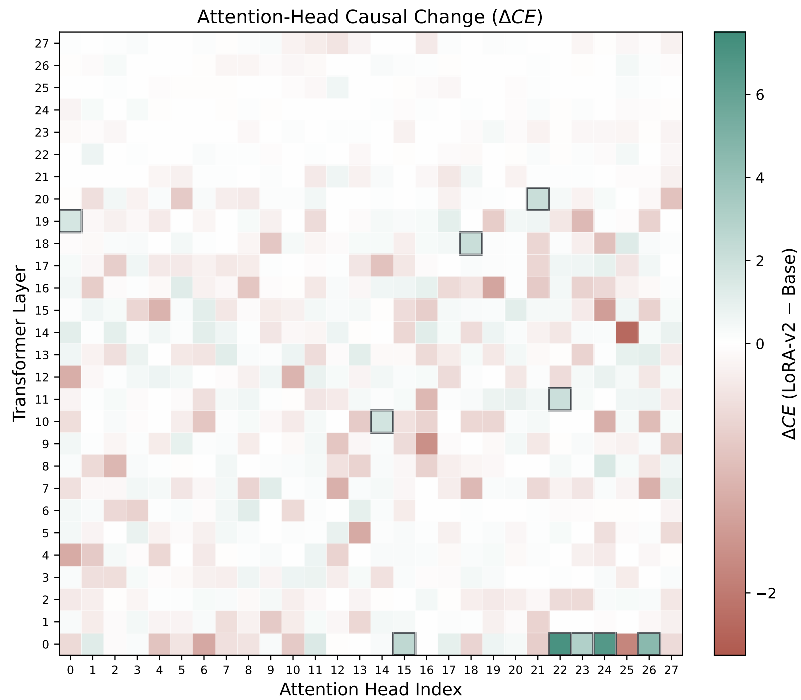


Figure 5: Attention-head heatmap of Base-to-LoRA causal change ΔCE for Qwen2.5-7B. Rows correspond to Transformer layers and columns correspond to attention-head indices. The positive-direction hue indicates $\Delta CE > 0$, whereas the negative-direction hue indicates $\Delta CE < 0$; the same directional color semantics are used throughout the paper. Black outlines denote the ten heads with the largest positive causal changes. The distribution shows sparse and heterogeneous redistribution of causal dependence across attention heads.

However, high positive ΔCE values are not restricted to the earliest layer. Heads in Layers 10, 11, 18, 19, and 20 also appear among the ten highest-ranked components.

The mechanism therefore cannot be characterized as limited to early layers. Rather, the strongest individual causal amplification is concentrated in several early heads while additional high-impact components remain distributed across intermediate layers.

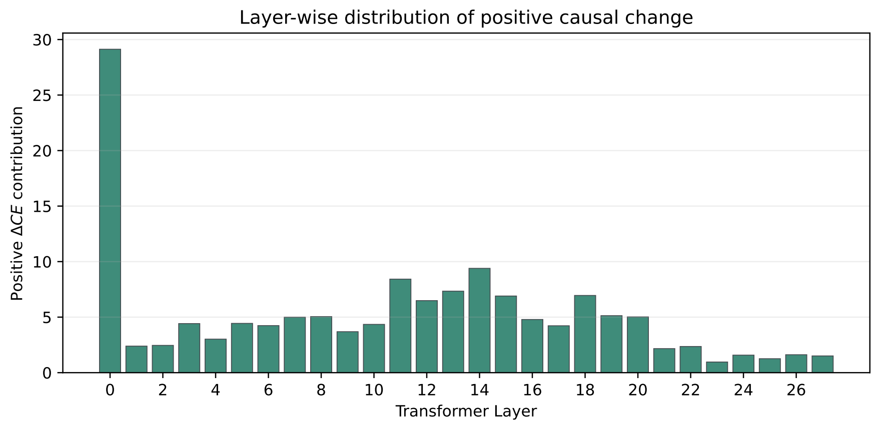


Figure 6: Layer-wise distribution of positive causal contribution measured by ΔCE across Qwen2.5-7B attention heads. The distribution shows non-uniform changes in causal contribution across Transformer layers after fine-tuning.

5.3.3. Representative Head: L0-H22

L0-H22 provides the clearest example of causal amplification after fine-tuning. In the Base model, ablating this head reduces the preference-shift score from 1.390711 to 0.751787, giving $CE_{\text{Base}} = 0.638924$. In LoRA-v2, the same ablation changes the score from 4.371395 to -3.329531, giving $CE_{\text{LoRA}} = 7.700926$. The resulting Base-to-LoRA change is therefore $\Delta CE = 7.062002$.

L0-H22 is thus already positively involved in the measured behavior before fine-tuning, but its causal contribution becomes much larger afterward. This pattern is consistent with selective amplification or functional reconfiguration of an existing component rather than the appearance of an entirely new attention-head function.

5.3.4. Causal Heterogeneity Across Heads

The complete scan reveals several qualitatively different patterns of causal change. Some heads satisfy $CE_{\text{LoRA}} \gg CE_{\text{Base}} > 0$, indicating strong amplification of an already positive Base-model contribution, while others have weak Base contributions but become substantially positive after fine-tuning.

Table 6: Top ten Qwen2.5-7B attention heads ranked by Base-to-LoRA causal change ΔCE.

Rank	Attention Head	ΔCE
1	L0-H22	7.062
2	L0-H24	6.671
3	L0-H26	4.552
4	L0-H23	2.996
5	L0-H15	2.486
6	L20-H21	2.101
7	L18-H18	2.055
8	L11-H22	2.046
9	L10-H14	1.746
10	L19-H0	1.595

Many heads remain near $\Delta CE \approx 0$, whereas a smaller subset has $\Delta CE < 0$, showing that fine-tuning can also reduce the measured contribution of individual components.

These results support a sparse and non-uniform redistribution of causal dependence rather than uniform strengthening of attention heads across the network.

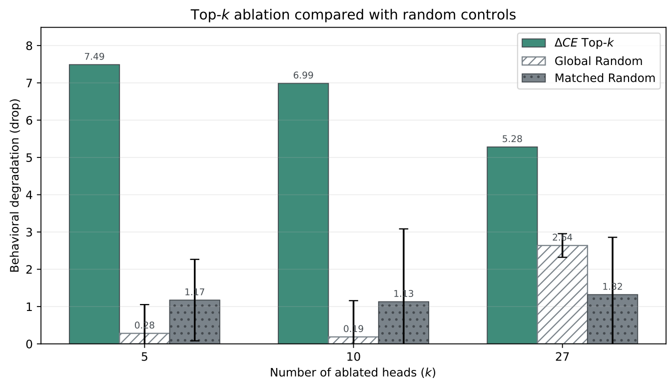
They also explain why attention-based correlation alone is insufficient for identifying the relevant components: strong causal importance after fine-tuning does not necessarily correspond to a simple observable attention–behavior correlation.

5.3.5. Top-k Joint-Ablation Validation

Single-head causal effects identify components whose individual importance changes after fine-tuning, but they do not establish whether the highest-ranked heads form a collectively important functional subset.

We therefore jointly ablate the Top-k heads ranked by ΔCE for $k \in \{5, 10, 27\}$ and compare the resulting behavioral degradation against Global Random and Matched Random controls, each averaged over five independently sampled control sets.

The causal ablation comparison is summarized in Figure 7.



k	Top-k	Global Random	Matched Random
5	7.489	0.281	1.173
10	6.987	0.186	1.130
27	5.281	2.638	1.320

Figure 7: Causal validation of Delta$_{CE}$-selected attention heads. Top-k ablation produces substantially larger behavioral degradation than Global Random and Matched Random controls, indicating that the identified heads form a functionally distinctive subset rather than arbitrary or layer-dependent components. Error bars indicate standard deviation across five random control runs.

Across all three intervention sizes, Top-k ablation causes greater behavioral degradation than either random control.

At $k = 5$, removing the five highest-ranked heads produces a drop of 7.489, compared with 0.281 under Global Random and 1.173 under Matched Random.

At $k = 10$, the Top-k drop remains large at 6.987, whereas Global Random and Matched Random produce drops of only 0.186 and 1.130, respectively.

At $k = 27$, Top-k ablation produces a drop of 5.281. Although the Global Random effect becomes larger at this intervention size, reaching 2.638, it remains substantially below the Top-k effect. The corresponding Matched Random drop is 1.320.

These results provide stronger evidence that the ΔCE ranking identifies a functionally distinctive subset of attention heads.

The Global Random comparison shows that the result cannot be explained simply by removing the same number of arbitrary attention heads.

The Matched Random comparison provides a stricter control by preserving the layer distribution of the Top-k set. The persistence of a larger Top-k effect under this control indicates that the result also cannot be explained solely by the network depths at which the selected heads are located.

5.3.6. Non-Monotonic Joint-Ablation Effects

An important feature of the Top-k results is that behavioral degradation does not increase monotonically with k: the Top-k drops are 7.489, 6.987, and 5.281 for $k = 5$, 10, and 27, respectively. Removing additional high-ranked heads therefore does not produce progressively larger degradation.

This pattern is consistent with the non-additive formulation in Eq. 12 and may reflect redundancy, compensation, inhibitory interactions, or other nonlinear dependencies among attention heads.

The present experiment does not distinguish among these possibilities. Nevertheless, it shows that a relatively small set of highly ranked heads can have a particularly strong joint effect and that extending the intervention to a larger set changes the functional interaction among the remaining components.

Overall, the causal analyses show a sparse and heterogeneous redistribution of head-level dependence after fine-tuning. The highest-ranked heads form a functionally distinctive subset under joint ablation, but these interventions directly explain BCDM-based preference adaptation rather than the mechanism of ACL Bench transfer.

6. Discussion

The results indicate that rule-constrained fine-tuning produces selective rather than uniform changes. All three model families improve in overall

ACL Bench accuracy, but the category-level effects differ substantially. For Qwen2.5-7B, the same fine-tuning also strengthens constraint-sensitive preference reallocation and changes the causal contribution of a limited subset of attention heads. These effects are related, but they should not be collapsed into a single claim of general capability improvement.

6.1. Selective and Model-Dependent Transfer

The ACL Bench results show different transfer profiles across the three model families. Qwen2.5-7B improves strongly on Hierarchy Logic and Pattern Recognition while declining on Counter Logic. Llama3.1-8B gains most on Pattern Recognition, whereas the DeepSeek model shows smaller but more broadly distributed improvements.

This pattern argues against interpreting fine-tuning as a task-independent increase in reasoning ability. A more plausible interpretation is that training strengthens some operations that are repeatedly exercised in the source domain and remain useful after the surface form changes. The extent of transfer then depends on both the base model and the structure of the target task.

We therefore describe the observed effect as *selective structural transfer*. The relevant unit of transfer is not necessarily the source task as a whole, but reusable operations such as ordered comparison, pattern processing, and conditional action selection. The Qwen2.5-7B Counter Logic decrease is important in this respect: a model can improve overall while losing performance in a specific reasoning category. Category-level analysis is therefore necessary for distinguishing transfer from a uniform capability gain.

6.2. Constraint Learning as Preference Reallocation

The constraint-sensitive probes clarify the behavioral change learned by LoRA-v2. The best constraint delta margin measures whether an explicit prohibition shifts preference away from the forbidden action and toward the strongest legal alternative. This is more informative than measuring suppression of the prohibited action alone, because reduced probability does not guarantee that probability mass moves toward a valid alternative.

The substantially larger mean BCDM after fine-tuning indicates stronger sensitivity to changes in the admissible action space. The constraint-injected examples differ from ordinary state–action supervision because the appropriate output depends on an additional rule: the same underlying state can require a different action once a preferred option becomes invalid. Repeated exposure to this structure may therefore encourage conditional decision policies rather than fixed state–action mappings.

This interpretation remains limited by the current comparison. The main experiments contrast Base with LoRA-v2, so the incremental contribution of constraint-specific examples is not fully isolated from the broader effect of domain fine-tuning. A strictly matched LoRA-v1 versus LoRA-v2 comparison would be needed to establish that distinction directly.

6.3. Sparse and Non-Additive Causal Reorganization

The complete Qwen2.5-7B head scan reveals changes in causal dependence unevenly across attention heads. Most heads exhibit relatively small Base-to-LoRA changes, whereas a much smaller subset shows large positive ΔCE values and some heads show negative changes. This distribution is inconsistent with uniform strengthening of the attention system and instead supports a selective reorganization of functional dependence.

L0-H22 provides a representative example. The head contributes positively to the measured behavior in the Base model and becomes substantially more important after fine-tuning. This is consistent with prior findings that fine-tuning can amplify mechanisms already present in a pretrained model rather than necessarily constructing entirely new computational structures [15]. The present evidence supports this interpretation for some heads, but not as a universal account of every causal change.

The Top-k experiments provide a stronger test than single-head ranking alone. Joint ablation of the highest-ranked heads produces greater degradation than both global-random and layer-matched random controls, showing that the selected head identities matter beyond intervention size or layer placement. At the same time, degradation is non-monotonic as k increases. Individual head effects therefore cannot be treated as additive; redundancy, compensation, inhibitory interactions, or larger network-state changes may contribute to the joint-ablation pattern.

The exploratory attention-based screening also differs from the causal results. More generally, prior work cautions against treating attention-based signals as faithful explanations of model behavior [25]. The absence of LoRA-v2 heads surviving the correlation-based screening therefore does not imply an absence of functionally important heads, because causal ablation identifies several with large behavioral effects. Attention correlation and causal contribution should be treated as distinct quantities.

Several high-ΔCE heads occur in early layers, but the strongest effects are not confined there. High-impact heads also appear at intermediate depths, and the layer-matched controls show that the Top-k effect cannot be reduced to generic sensitivity of particular layers. The causal redistribution is better characterized as sparse and distributed across depth.

6.4. Scope, Limitations, and Causal Boundary

The behavioral and mechanistic results should be interpreted at different levels. ACL Bench provides evidence of selective cross-task transfer. The constraint probes measure stronger preference reallocation under explicit prohibitions. The attention-head interventions establish causal support for that constraint-sensitive preference shift. The current experiments do not directly show that the same heads are responsible for ACL Bench gains.

This distinction is important because the mechanistic analysis is evaluated with BCDM rather than ACL Bench. A direct causal bridge would require evaluating ACL Bench under the same Top-k and layer-matched interventions. Stronger ACL degradation under Top-k ablation would support a shared internal basis between constraint adaptation and transfer; stable ACL performance would instead suggest partially distinct mechanisms.

Several additional limitations remain. Dou Dizhu is the only source domain, and ACL Bench is intentionally constructed around structurally related operations, so broader generalization requires additional source domains and independent benchmarks. The primary training results are based on individual fine-tuned checkpoints rather than repeated training runs across multiple random seeds. Complete causal analysis is currently available only for Qwen2.5-7B, and Top-k ablation identifies a functionally important subset rather than a complete circuit.

The most informative extensions are a matched LoRA-v1/LoRA-v2 comparison, direct ACL Bench ablation with Top-k and matched controls, and causal replication on additional model families. Interaction-level analysis among the highest-ranked heads would also help distinguish redundancy, compensation, and other sources of the observed non-monotonic joint effects.

7. Conclusion

This study examines how rule-constrained fine-tuning affects cross-task transfer, constraint-sensitive preference reallocation, and internal causal dependence. Using Dou Dizhu as a structured source domain, we evaluate three open-weight models on ACL Bench and find higher overall accuracy after fine-tuning, with strongly model- and category-dependent effects. For Qwen2.5-7B, the best constraint delta margin also increases substantially, indicating stronger reallocation of preference from prohibited actions toward legal alternatives.

Causal attention-head ablation further shows that these preference shifts depend disproportionately on a small subset of heads rather than on uniform changes across the attention system. Top-k joint ablation produces greater degradation than both global-random and layer-matched controls, while the non-monotonic effects indicate that head contributions are not simply additive. These results support a view of fine-tuning as selective behavioral and mechanistic reorganization rather than uniform capability gain. The causal evidence currently concerns constraint-sensitive preference behavior; whether the same components also mediate ACL Bench transfer remains an open question.

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